

Enabling large-scale underground hydrogen storage in aquifer-driven reservoirs: Integrated technology and economic feasibility

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HIGHLIGHTS

- Investigates the feasibility and advantages of large-scale UHS in aquifer-driven reservoirs.
- Identifies the relative permeability and capillary force among CH₄, N₂, and CO₂ for cross-comparison.
- Develops a real-world ultra-fine UHS model that considers the difference in flow ability among cushion gases.
- Different cushion gases show significant variability in their effectiveness at preventing water intrusion.
- Aquifer-driven reservoirs can enable profitable, large-scale UHS at prevailing natural gas storage costs.

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ABSTRACT

Approximately 30%–55% of existing gas reservoirs are aquifer-driven. Converting this subset for Underground Hydrogen Storage (UHS) could help maintain production pressure, provide peak-shaving capacity, and reduce hydrogen costs. However, these reservoirs also face a significant risk of water invasion when repurposed for UHS. In this study, we develop an integrated workflow that combines pore network modeling, numerical simulation, and an economic assessment based on NPV and IRR. Unlike conventional simulation studies, our workflow captures differences in subsurface mobility between hydrogen and various cushion gases. We apply this workflow to evaluate the production performance and cost effectiveness of a real world depleted aquifer driven reservoir using its field data. The results show that various cushion gases differ in their ability to resist water intrusion in aquifer driven reservoirs; CO₂ achieves the highest hydrogen recovery and the most favorable economic performance. Furthermore, the storage cost of UHS can be reduced to one tenth of that for current surface storage methods. This finding suggests that, with appropriate cushion gas selection, aquifer driven reservoirs hold substantial potential for conversion to UHS and carry important implications for the future low cost utilization of hydrogen.

1. Introduction

The Paris Agreement has set the goal of limiting global warming to well below 2 °C and pursuing 1.5 °C [1]. As a result, nations have adopted successive net zero timetables [2]. Hydrogen is recognized as a

key energy vector for deep decarbonization, particularly in energy-intensive sectors such as peaking power generation, heavy-duty transport, and high-temperature industrial processes [3]. As shown in Fig. 1 (a), global hydrogen demand surged from approximately 97 Mt. in 2022 and is projected to exceed 150 Mt. by 2050 [4]. This trend reveals a

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growing temporal mismatch between renewable hydrogen supply and end-user demand, making secure, cost-effective, and large-scale Underground Hydrogen Storage (UHS) essential for achieving net-zero targets.

Currently, hydrogen storage and transport rely primarily on above-ground high-pressure tanks [8]. While these methods are well established, they incur high equipment and land costs, resulting in storage costs of approximately USD 12.7 / kg, which are unlikely to support large-scale hydrogen turnover. In contrast, as shown in Fig. 1(b), UHS reduces the unit storage cost for a three-month period to approximately USD 0.7 / kg, representing an order-of-magnitude cost reduction [5,6]. In addition, UHS allows continuous operation of electrolyzers in green hydrogen systems, lowering equipment depreciation and mitigating electricity price variability. As shown in Fig. 1(c), the hydrogen production cost decreases with increasing annual operating hours, and this trend is more pronounced under high electricity price scenarios [7]. Consequently, UHS is widely regarded as a preferred option for peak hydrogen supply, low-cost production, and flexible system operation [9].

Evidence from field compositional data, macroscopic gas flow calculations, and molecular simulations indicates that the caprocks of most water-driven depleted gas reservoirs can provide effective sealing for hydrogen [6]. Field observations show that hydrogen is commonly preserved in natural gas reservoirs at low molar fractions (typically below 2%) over geological timescales, suggesting long-term containment by caprocks. At the reservoir scale, leakage calculations show that annual hydrogen leakage rates remain low, on the order of 3–100 kg/m², partly due to the relatively high capillary pressure of hydrogen [10]. At the pore scale, molecular simulations show that water clustering creates high-energy barrier layers that strongly limit hydrogen leakage in pores smaller than ~ 8.5 Å [11]. Consistent with these findings, several European pilot projects, including Hydrogen Kinsale, Storage Hub Italy, and Humber H₂ Storage, have adopted depleted gas reservoirs with demonstrated operational safety and acceptable recovery [7,12].

Over the past sixty years, Underground Gas Storage (UGS) practice has confirmed the high porosity and permeability of depleted gas reservoirs, confirming their suitability as a reliable medium for large-scale energy storage [13]. For example, over 75% of the more than 400 UGS facilities in the United States are in depleted gas reservoirs, and their working gas volumes substantially exceed those of salt caverns and aquifer-based facilities, as shown in Fig. 2(a) to Fig. 2(c) [14]. Moreover,

aquifer-driven reservoirs represent a significant subset of these sites, accounting for 30–55% of depleted gas reservoirs across different regions. As shown in Fig. 2(d), aquifer-driven reservoirs account for 54%, 45%, 34%, and 32% of depleted gas reservoirs in China, the United States, the EU, and Canada, respectively. Accordingly, aquifer-driven reservoirs represent a substantial and underutilized resource for large-scale UHS development.

Unlike closed reservoirs, aquifer-driven reservoirs offer specific geological advantages when converted into UHS, as shown in Fig. 3. First, continuous recharge from the surrounding aquifer maintains a gradual pressure decline and stable bottom-hole pressure [15,16]. Furthermore, repeated injection–production cycles can drive the aquifer boundary outward, thereby expanding the mobilizable storage capacity [17]. Field data from natural gas storage facilities indicate that this expansion can reach 5–15%. In addition, aquifer-driven reservoirs typically feature high porosity and permeability, enabling efficient injection and withdrawal at lower storage costs [18].

However, high flow connectivity between a storage reservoir and an aquifer substantially increases the risk of water intrusion. In a UHS case study from New Zealand, Liu found that cyclic injection–production operations can drive upward migration of a gas–water contact and thus increase water intrusion [19]. Liao further reported that water intrusion can occur even in weak aquifers, because repeated pressure perturbations can promote gradual water ingress into a storage reservoir [20]. Furthermore, Jiang examined the mechanisms and found that capillary forces largely controlled water movement; with additional operation cycles, residual water gradually became isolated in clusters or became trapped in pore throats [21].

As a result, the importance of cushion gas has been increasingly recognized in recent studies. Kanaani showed that cushion gas can mitigate gravity segregation and thereby limit the upward migration of hydrogen [22]. Saeed reported that gravity overriding, viscous fingering, and lateral spreading are three primary controlling mechanisms in UHS [23]. Furthermore, Zhao found that a fluid density was a key governing parameter, and a higher density can enhance the recovery factor of UHS [18].

Recent studies have shown that accurate representation of gas mobility is critical in numerical simulations. For example, inaccurate relative permeability curves can introduce errors exceeding 141% [24]. However, obtaining accurate curves is challenging. Few laboratories have conducted relative permeability tests for H₂ due to the requirement

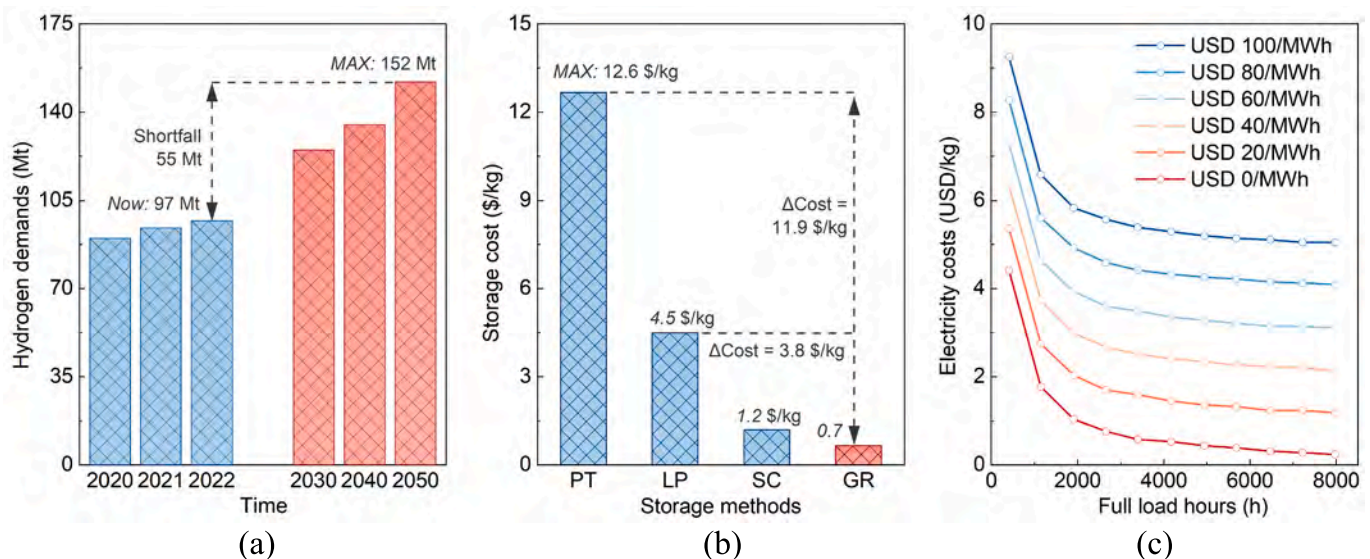


Fig. 1. Global hydrogen demand and corresponding storage and production costs. (a) Global demand for hydrogen forecasts [4]; (b) The current three-month storage cost of hydrogen under different methods [5,6]; (c) Variation in hydrogen production costs [7]. Where PT refers to above ground Pressurized Tank, LP refers to Line Packing, SC refers to underground Salt Cavern, and GR refers to underground Gas Reservoir.

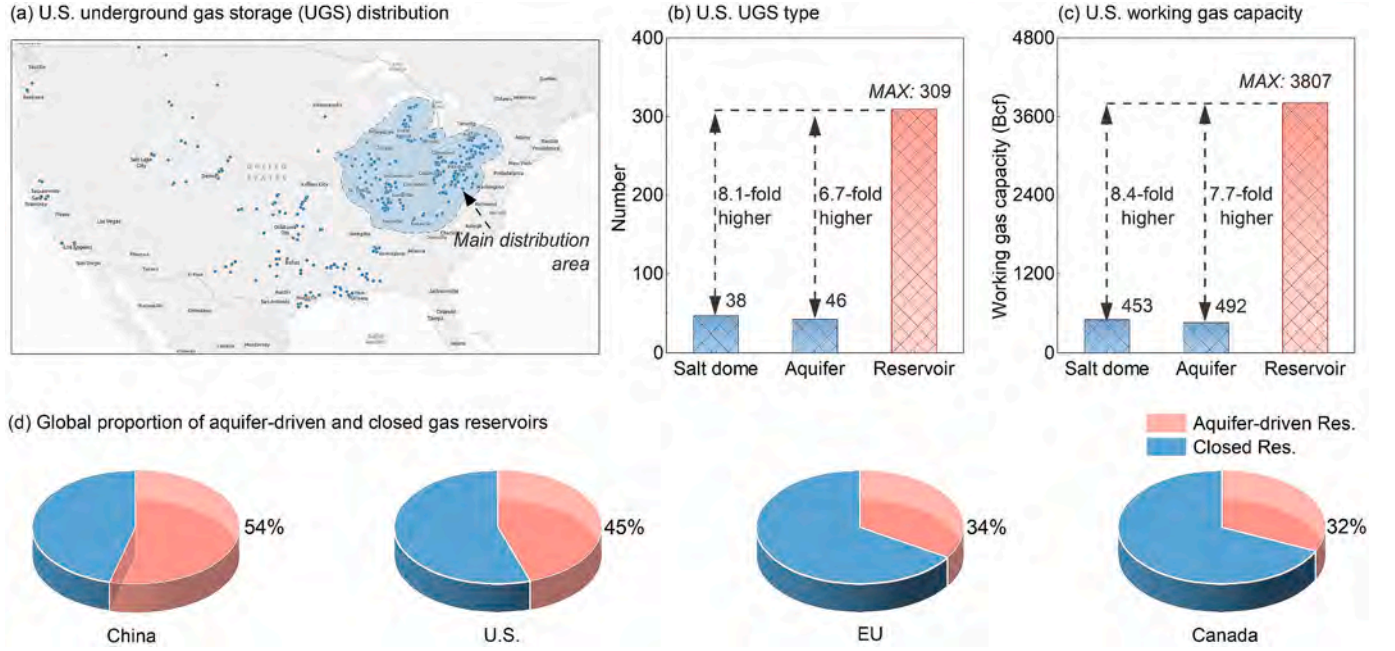


Fig. 2. UGS demand and storage types. UGS demand and storage types. (a) Distribution of UGS facilities in the U.S. (b) Number of UGS sites by type. (c) Working gas capacity by storage type. (d) Shares of aquifer-driven and closed reservoirs in major regions. Where Res. refers to reservoirs.

for specialized explosion-proof equipment [25]. Consequently, finding relative permeability curves for hydrogen and cushion gas in the same rock core is very difficult. Therefore, current numerical simulation studies often neglect differences in relative permeability curves among gases, resulting in significant discrepancies between simulated and actual cushion gas performance in UHS.

To evaluate cushion gas performance in aquifer-driven reservoirs converted to UHS, this study explicitly addresses gas-dependent differences in relative permeability and capillary forces that have been overlooked in previous work. A pore network model (PNM) is used to quantify relative permeability and capillary force variations among CH_4 , N_2 , and CO_2 , which are then incorporated into a reservoir-scale numerical model to assess their impact on large-scale UHS operation. Field data from a depleted gas reservoir are further integrated to evaluate UHS economic performance. Section 2 outlines the methodology, including the PNM and compositional modeling framework. Section 3 presents simulation results and the economic assessment. Section 4 discusses key technical insights and limitations, and Section 5 summarizes the main conclusions.

2. Methodology

This study proposes a workflow to couple gas-specific flow properties with UHS performance and economics, as shown in Fig. 4. Fluid properties, including density, viscosity, and interfacial tension, are first obtained from measured data of the studied reservoir. For fluid-property and phase-behavior calculations, the Peng–Robinson Equation of State (PR-EOS) is applied. A PNM is then used to quantify gas-dependent relative permeability and capillary forces. These parameters are then embedded into a compositional reservoir model to evaluate cushion-gas performance. Compared with conventional studies using uniform parameters, this workflow enables more reliable techno-economic assessment.

2.1. Pore network model

The PNM represents the reservoir as a statistically equivalent pore–throat network, to characterize capillary pressure and relative

permeability during multiphase flow. The pore and throat size distributions and network connectivity were calibrated to match the representative porosity and permeability of the target reservoir. It is mainly used to quantify the influence of gas-property differences on relative permeability and capillary pressure.

In pore network modeling, flow is computed using the Hagen–Poiseuille equation, as shown in Eq. 1, where hydraulic conductance and the size factor F represent pore–throat geometry [26]:

$$Q = \frac{10^8 \pi R_{I-J}^4 (p_I - p_J)}{8 \mu L_{I-J}} \quad (1)$$

$$\gamma_g = \frac{10^3 F}{\mu} = \frac{10^3 \pi r_{I-J}^4}{8 \mu L_{I-J}} \quad (2)$$

The determination of the overall conductance of the pore–throat–pore conduit involves summing the conductance values of three components. The hydraulic conductance between pores i and j , and throat k , can be calculated using the following Eq. [27]:

$$Q = 10^7 \left(\frac{\mu}{F_I} + \frac{\mu}{F_K} + \frac{\mu}{F_J} \right)^{-1} (p_I - p_J) \quad (3)$$

The capillary force are calculated using:

$$p_c = p_{nw} - p_w \quad (4)$$

Relative permeability can be calculated using the following equation:

$$k_r = \frac{k_e}{k_{abs}} = \frac{Q_e}{Q_{abs}} = \frac{u_e}{u_{abs}} \quad (5)$$

where Q is the volumetric flow rate through the pore–throat–pore conduit [m^3/s]; R_{I-J} is the effective radius of a pore or throat between pores I and J [m]; r_{I-J} is the radius of the throat connecting pores I and J [m]; L_{I-J} is the length of the throat between pores I and J [m]; p_I and p_J are the pressures at pores I and J , respectively [Pa]; μ is the dynamic viscosity of the fluid [$\text{Pa}\cdot\text{s}$]; γ_g is the hydraulic conductance of the pore or throat [$\text{m}^3\cdot\text{Pa}^{-1}\cdot\text{s}^{-1}$]; F is the geometrical size factor of the pore or throat [m^4]; F_I , F_K , and F_J are the size factors of pore I , throat K , and pore J , respectively [m^4]; p_c is the capillary pressure [Pa]; p_{nw} and p_w are the

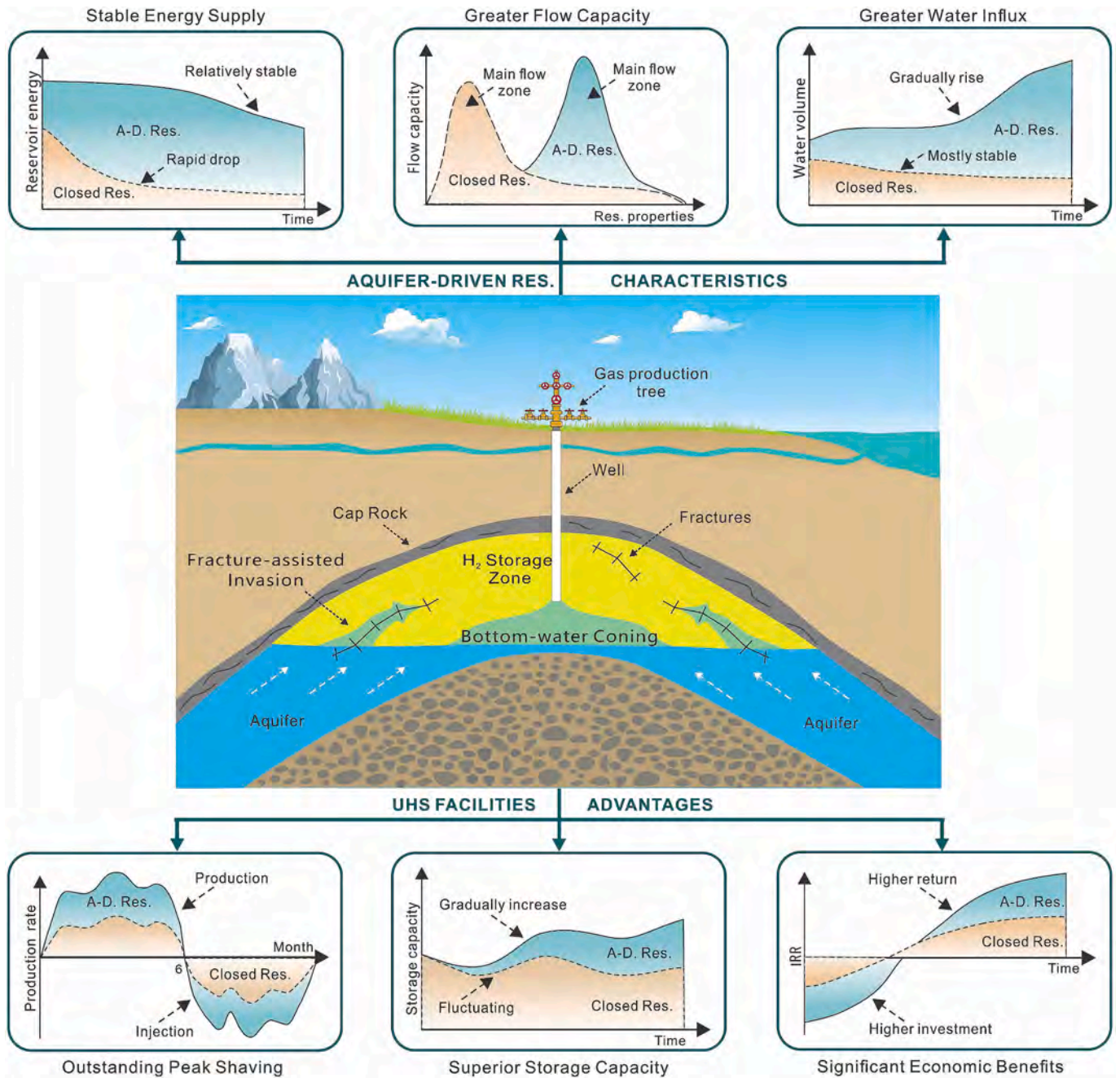


Fig. 3. Schematic of the aquifer-driven UHS facility: top panels depict key reservoir characteristics; bottom panels show corresponding facility advantages. Where A-D. Res. refers to aquifer-driven reservoir.

pressures of the non-wetting phase and wetting phase, respectively [Pa]; k_r is the relative permeability [dimensionless]; k_e is the effective permeability under multiphase flow conditions [m^2]; k_{abs} is the absolute permeability of the porous medium [m^2]; Q_e and Q_{abs} are the volumetric flow rates under effective and single-phase conditions, respectively [m^3/s]; u_e and u_{abs} are the Darcy velocities under effective and absolute permeability conditions, respectively [m/s].

In this study, mixed-gas effects are represented using a dynamic interpolation approach. Pure-gas relative permeability and capillary pressure curves are first generated with the PNM. The numerical model then computes the local gas composition in each grid cell, which is used to interpolate among these pure-gas curves, yielding an equivalent relative permeability and capillary pressure that vary dynamically in space and time.

2.2. Numerical simulation

After completing the pore-scale parameter characterization (Section 2.1, Pore Network Model), the resulting relative permeability and capillary pressure curves are incorporated into the compositional numerical simulation framework (Section 2.2, Numerical Simulation) to characterize multiphase flow behavior at the reservoir scale.

Regarding the numerical implementation, the UHS process is solved using a fully implicit compositional simulation approach (Section 2.2.1, Component model). The governing equations consist of the component mass conservation equations for both the liquid and gas phases (Eqs. 6–7), the Darcy flow equations (Eq. 8), and the thermodynamic phase equilibrium constraints based on an equation of state (Eq. 9). These equations are assembled into a unified system and solved

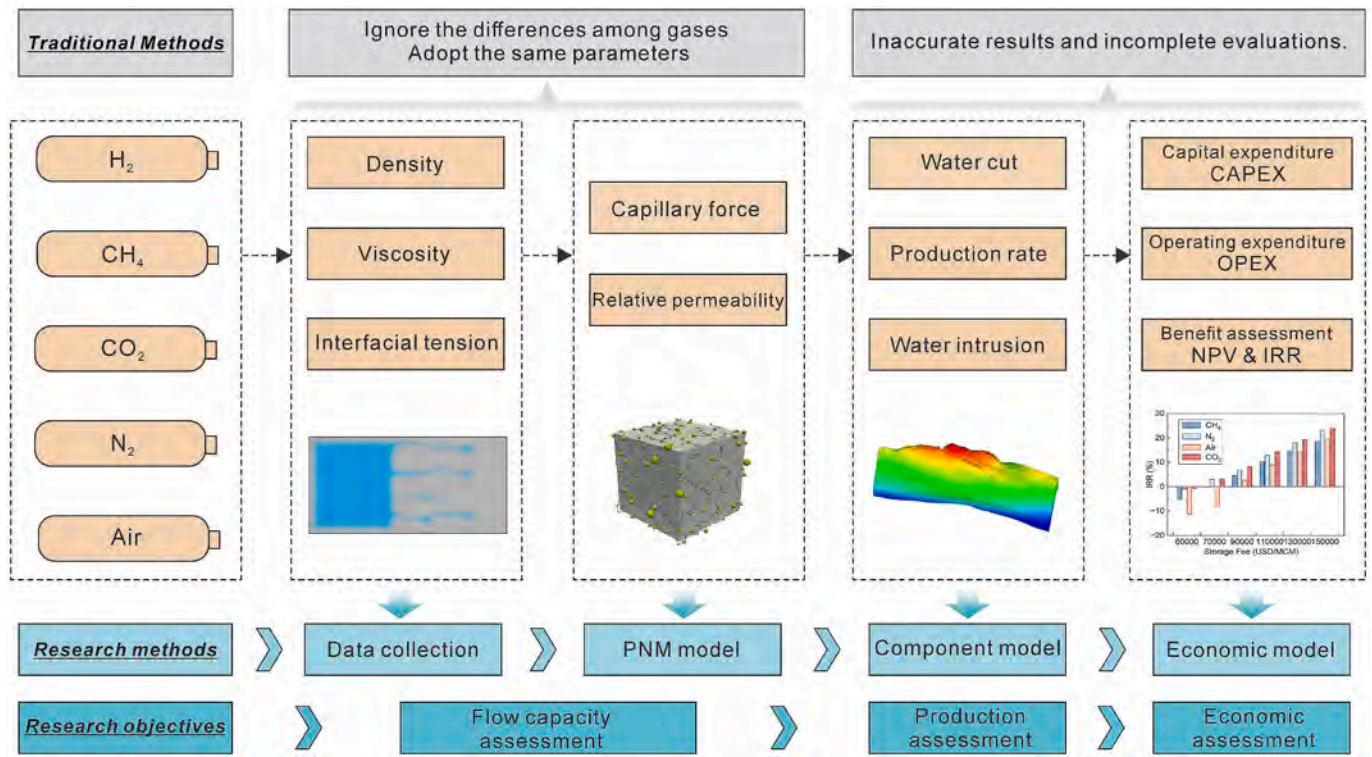


Fig. 4. The technical workflow of this study and comparison with traditional research methods.

simultaneously at each time step. The primary unknowns include pressure, phase saturations, and the molar fractions of each component in the different phases.

Within a given time step, fugacity coefficients are first calculated using the PR-EOS based on the current pressure, temperature, and component composition (Section 2.2.2, Fluid model). Gas–liquid phase equilibrium calculations are then performed by enforcing fugacity equality to determine phase compositions and the corresponding thermophysical properties. These updated properties are subsequently substituted into the component mass conservation equations and the flow equations, resulting in a nonlinear system of equations. The nonlinear system is solved using the Newton–Raphson iterative method until convergence of the primary variables is achieved within predefined tolerance criteria. Time advancement is handled using an implicit discretization scheme to ensure numerical stability and computational robustness under multicomponent, multiphase flow conditions.

Reservoir-scale simulations were conducted using the compositional simulator in tNavigator, based on a fully implicit compositional formulation that satisfies mass conservation and local thermodynamic equilibrium under Darcy's flow. Gas–liquid phase behavior is described by an equation of state, while relative permeability and capillary pressure under gas-mixture conditions are obtained through dynamic interpolations of the pure-gas curves according to the grid-scale component fractions. This model is suitable for comparing the relative impacts of different cushion-gas strategies on water-invasion control and hydrogen recovery under fixed geological conditions.

2.2.1. Component model

The component model is based on a compositional framework that couples component mass conservation with Darcy's flow and phase equilibrium. Fluid properties and phase behavior are updated via an equation of state using fugacity equality, enabling simultaneous solution of pressure, saturations, and compositions [28,29].

Water component:

$$-\nabla \cdot (\xi_w u_w) = \frac{\partial}{\partial t} (\phi s_w \xi_w) + Q_w \quad (6)$$

i -th component:

$$-\nabla \cdot (x_i \xi_l u_l + y_i \xi_g u_g) = \frac{\partial}{\partial t} (\phi (s_l x_i \xi_l + s_g y_i \xi_g)) + Q_i \quad (7)$$

Darcy's equation:

$$u = -\frac{10^{-7} k k_r}{\mu} [\nabla p - \rho g \nabla Z] \quad (8)$$

The EOS is used to express instantaneous thermodynamic phase equilibrium, based on the concept of equal fugacity between liquid and gas phases [30]:

$$f_{i,l}(p, T, x_1, x_2, \dots, x_{nc}) = f_{i,g}(p, T, y_1, y_2, \dots, y_{nc}) \quad (9)$$

where ϕ is the porosity [dimensionless]; s_w and s_g are the saturations of the water phase and gas phase, respectively [dimensionless]; ξ_w is the molar density of the water component [mol/m³]; ξ_l and ξ_g are the molar densities of the liquid phase and gas phase, respectively [mol/m³]; x_i and y_i are the mole fractions of the i -th component in the liquid phase and gas phase, respectively [dimensionless]; u_w , u_l , and u_g are the Darcy velocities of the water phase, liquid phase, and gas phase, respectively [m/s]; Q_w is the source/sink term for the water component [mol/(m³·s)]; Q_i is the source/sink term for the i -th component [mol/(m³·s)]; u is the Darcy velocity [m/s]; k is the absolute permeability of the reservoir [m²]; p is the pressure [Pa]; ρ is the mass density of the fluid [kg/m³]; g is the gravitational acceleration constant [m/s²]; Z is the vertical depth measured positively upward [m]; $f_{i,l}$ and $f_{i,g}$ are the fugacities in the liquid phase and gas phase, respectively [Pa]; T is the temperature [K]; $x_1, x_2, \dots, x_m$ and $y_1, y_2, \dots, y_m$ are the mole fractions of all m components in the liquid and gas phases, respectively [dimensionless].

2.2.2. Fluid model

In this study, fluid thermodynamic behavior and phase equilibrium are described using a compositional formulation based on the PR-EOS. The equation of state is employed to couple pressure, temperature, and fluid composition and to evaluate phase behavior under UHS conditions.

At each simulation time step, gas–liquid phase equilibrium is determined through an isothermal flash calculation constrained by the equality of fugacities between the gas and liquid phases. For a given pressure, temperature, and overall component composition, the equation of state is solved together with material balance equations to determine the phase fractions and phase compositions. Within this framework, the dissolution of gas components into the water phase is represented implicitly as part of the thermodynamic equilibrium calculations: each component partitions between the gas and liquid phases according to the fugacity-equality condition, without introducing explicit solubility or kinetic dissolution models. Accordingly, H_2 is allowed to partition into the aqueous phase, and its dissolved amount is determined self-consistently during the flash calculations together with the other components.

As a result, gas dissolution in the aqueous phase is determined self-consistently during flash calculations, ensuring thermodynamic consistency between phase behavior and compositional transport. The PR-EOS is expressed by the following core equation, where the mixture parameters $(a\alpha)_m$ and b_m are determined from the properties of the individual components [31,32]:

$$P = \frac{(1.80T + 491.67)R}{14.5038(V - b_m)} - \frac{(a\alpha)_m}{14.5038(V^2 + 2b_mV - b_m^2)} \quad (10)$$

Liquid and vapor phase composition:

$$x_i = \frac{c_i}{1 + \beta(K_i - 1)} \quad (11)$$

$$y_i = \frac{K_i c_i}{1 + \beta(K_i - 1)} \quad (12)$$

Molecular weights of the liquid and gas phases:

$$MW_l = \sum_{i=1}^{n_c} x_i MW_i \quad (13)$$

$$MW_g = \sum_{i=1}^{n_c} y_i MW_i \quad (14)$$

where V is the molar volume [m³/mol]; $(a\alpha)_m$ and b_m are the mixture attraction and covolume parameters, respectively [Pa·m⁶/mol²]; n_c is the total number of components [dimensionless]; c_i is the overall mole fraction [dimensionless]; β is the vapor-phase fraction [dimensionless]; K_i is the phase equilibrium ratio of component i , defined as the ratio of its mole fraction in the gas phase to that in the liquid phase [dimensionless]; MW_l and MW_g are the average molecular weights of the liquid and gas phases, respectively [kg/mol].

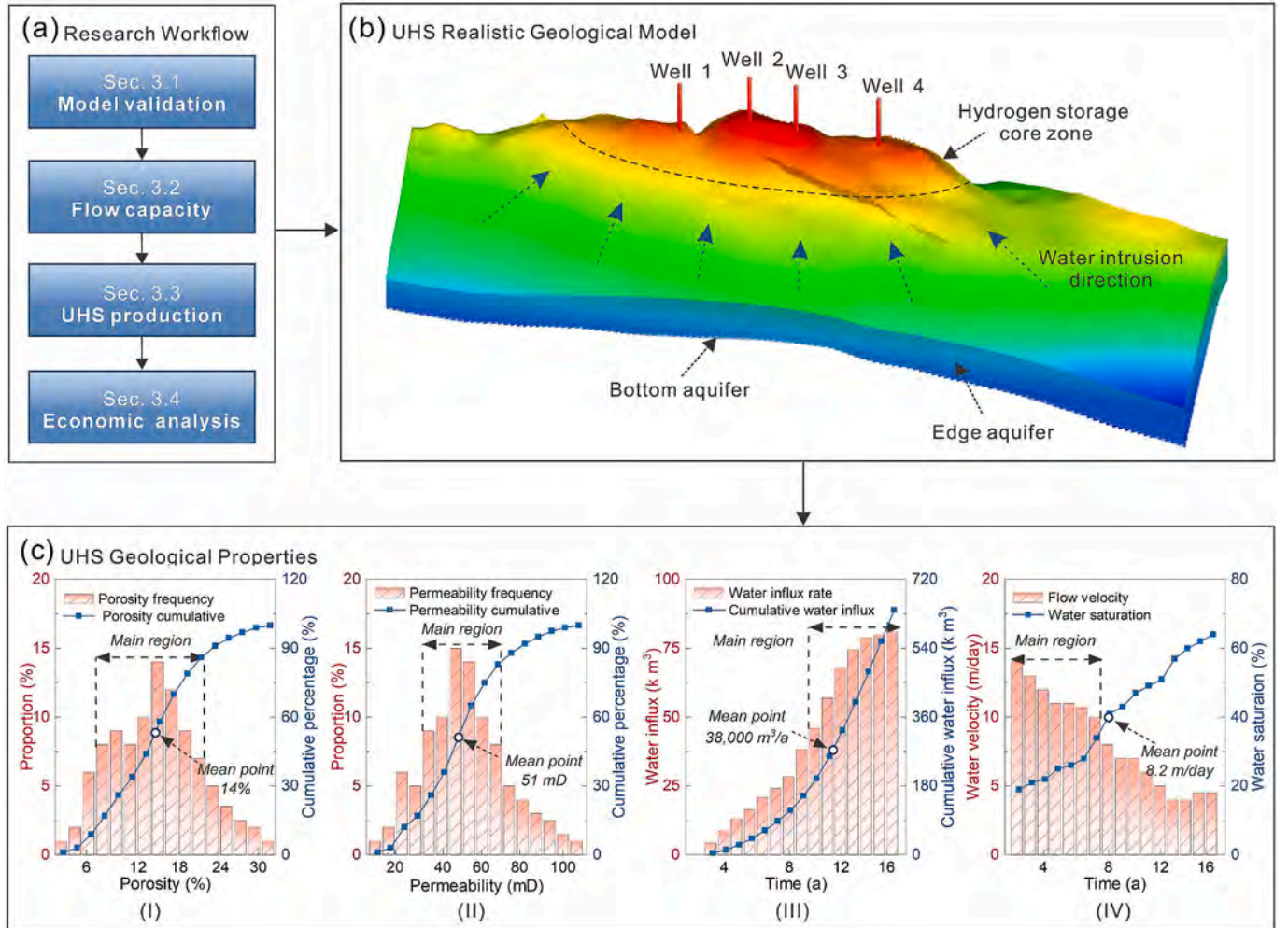


Fig. 5. Numerical simulation model of UHS used in this study. (a) Workflow of this study; (b) Geological model of gas reservoir A; (c) Reservoir properties of gas reservoir A.

Gas reservoir A is approximately 25–35 m thick and has been in production for nearly 16 years. Owing to increasing water intrusion and declining production rates, the site has ceased production and was selected as the test location for the UHS facility. The reservoir is a typical aquifer-driven system, connected to both bottom-water and edge-water aquifers, as shown in Fig. 5. It has an average porosity of 14.2% and an average permeability of 51.4 mD. The numerical model accurately reproduces the reservoir's geological features, using a horizontal grid resolution of 10 m and a vertical resolution of 0.1 m, for a total of 34.26 million grid cells.

2.2.3. Robustness of model

As the grid number increases, the size of the Jacobian matrix grows accordingly, leading to severe convergence challenges in large-scale numerical simulations. This issue is particularly pronounced in the UHS reservoir model used in this study, which contains more than 34.26 million grid blocks. To enhance numerical stability and computational efficiency, the grid structure, rock properties, and component representation were optimized. In addition, the PNM contributes to the overall robustness of the framework by providing gas-dependent relative permeability and capillary pressure curves for the compositional simulations.

First, extremely small grid blocks generated by unstructured gridding near fracture intersections were reduced by adjusting the local grid orientation around fractures, thereby improving solution stability. Second, extreme permeability and porosity values arising from kriging interpolation of well logs were smoothed. Although such values occur in only a limited number of cells and have negligible impact on flow behavior, they can significantly degrade Jacobian conditioning; smoothing therefore improves model robustness. Third, a pseudo-components approach was adopted to reduce model complexity. Based on acentric factor distributions, K-means clustering was applied to lump 15 original components into 7 pseudo-components (Table 1), reducing computational time by approximately 50%.

2.3. Economical model

2.3.1. Capital expenditure

Capital Expenditure (CAPEX) is a significant portion of UHS total expenses. As shown in Eq. 15, CAPEX comprises engineering construction, cushion gas, reservoir residual value, and construction contingency funds. Of these, engineering construction is the largest component,

Table 1
Pseudo-components lumping scheme [33].

Component	Acentric factor	K-means cluster center	Lumping scheme
N ₂	0.038	0.038	Component 1 (N ₂)
CO ₂	0.228	0.228	Component 2 (CO ₂)
C1	0.012	0.012	Component 3 (C1)
C2	0.100	0.126	Pseudo-component 1
C3	0.152		(C2-C3)
IC4	0.184	0.223	Pseudo-component 2 (C4-C6)
NC4	0.200		
IC5	0.228		
NC5	0.252		
C6	0.250	0.331	Pseudo-component 3 (C7-C10)
C7	0.280		
C8	0.312		
C9	0.348		
C10	0.385		
C11+	0.662	0.662	Pseudo-component 4 (C11+)

including drilling engineering, completion engineering, and surface engineering. Cushion gas costs depend on gas type and injection volume; both must be determined based on market prices and numerical simulation results to ensure an accurate estimate. Reservoir residual value refers to the remaining worth of the depleted gas field, including undeveloped natural gas resources and equipment reusable for UHS. The contingency fund ensures availability of resources for large-scale engineering construction.

$$CAPEX = C_{\text{construction}} + C_{\text{cushion gas}} + C_{\text{existing asset value}} + C_{\text{contingency fund}} \quad (15)$$

where CAPEX is the capital expenditure of this UHS project [USD]; $C_{\text{construction}}$ is the costs of UHS construction [USD]; $C_{\text{cushion gas}}$ is the costs of cushion gases [USD]; $C_{\text{existing asset value}}$ is the remaining worth of the depleted gas field [USD]; and $C_{\text{contingency fund}}$ is the temporary funds ensure the normal progress of UHS construction [USD].

2.3.2. Operating expenditure

Operating Expenditure (OPEX) accounts for a significant proportion of regular costs in UHS project operations. The OPEX model in this study includes production costs and indirect costs, as shown in Eq. 16. Production costs cover annual expenses for large-scale hydrogen injection and production, including storage facility operation fees such as staff salaries, electricity consumption, and wellbore inspections. Indirect costs primarily comprise management and sales expenses to ensure stable UHS operation.

$$OPEX = C_{\text{production costs}} + C_{\text{indirect expenses}} \quad (16)$$

where OPEX is the operating expenditure of the UHS project [USD]; $C_{\text{production costs}}$ is the costs of injection and operation [USD]; and $C_{\text{indirect expenses}}$ is the costs of management and sales [USD].

2.3.3. Assessment model

The Net Present Value (NPV) is a critical metric for assessing overall project profitability. The NPV model includes CAPEX, OPEX, taxes, and the discount rate, as shown in Eq. 17. The value of USD 126,923/MCM was used as the reference storage price for the base-case economic evaluation based on current market conditions, while a 10% discount rate was adopted. The impacts of key economic uncertainties were further examined through sensitivity analyses.

$$NPV = -CAPEX + \sum_{t=1}^N \frac{(C_{\text{revenue}} - OPEX - C_{\text{tax}})}{(1 + r_d)^t} \quad (17)$$

where NPV represents project's net present value [USD]; t is the year under calculation [a]; N the evaluation period for the entire project. This study assessed for 30 years [a]; C_{revenue} is the revenue obtained from hydrogen storage, with this model using natural gas storage fee as the reference [USD]; C_{tax} is the tax rate [%]; r_d represents the discount rate of the project[%].

The Internal Rate of Return (IRR) is another important project assessment goal, as shown in Eq. 18. While NPV is used to measure the absolute net economic benefit of a project over its lifetime, IRR reflects the rate of return and is useful for identifying the breakeven storage price. Thus, this study employs both NPV and IRR to evaluate the economic feasibility of hydrogen storage from complementary perspectives. In this study, the economic evaluation was based on base-case nominal cost and revenue inputs, with a discount rate of 10% adopted for the discounted cash-flow analysis.

$$NPV = \sum_{t=0}^N \frac{C_t}{(1 + IRR)^t} \quad (18)$$

where C_t represent the cash flow of the project at year t [USD].

3. Results and analysis

3.1. Model validation

Since Reservoir A has a 16-year production history, model accuracy can be verified by matching historical data with simulation results. Fig. 6 shows the relationship between simulated and historical production rates, with a mean absolute error (MAE) of 4.86% for cumulative gas production and 5.18% for cumulative water production. These results indicate that the model accurately reflects the reservoir's flow characteristics and can be applied to subsequent UHS development stages.

3.2. Flow capacity

A PNM was employed to derive gas-dependent capillary pressure and relative permeability curves in this study. The PNM adopts a pore-throat network representation, in which the porous medium is idealized as an interconnected system of pores and throats constructed in a statistical manner [27,34]. The pore and throat size distributions, as well as network connectivity, are parameterized to represent the characteristic pore structure of the studied reservoir rather than explicitly reconstructing the exact pore geometry.

The network consists of 4637 pores and 1476 throats, and its parameters were calibrated to match the reservoir permeability (51.4 mD) and porosity (14.2%). This calibration provides a statistically equivalent representation of the in-situ porous medium, enabling the generation of capillary pressure and relative permeability curves for CH₄, N₂, and CO₂ that consistently reflect gas-specific effects within the compositional simulation framework.

The relative velocity of different gases via water saturation in this process are shown in Fig. 7. According to the analysis, H₂ exhibits the lowest relative velocity, sustaining a velocity of less than 0.25 in most pores at 80% water saturation. CH₄ displays a slightly higher mean flow velocity than H₂, and it performs much better in well-connected pore channels, where it obtains the highest average flow velocity at 80% water saturation. CO₂ possesses the highest flow velocity among all the gases tested, outperforming other gases in both maximum and average velocities target at various water saturation levels.

3.2.1. Capillary pressure

For UHS, capillary pressure controls gas entry into water-saturated pores during injection and governs water re-invasion and gas release

during production, thereby influencing hydrogen retention and recoverability. During a drainage stage, gas must overcome the entry pressure to invade water-filled pores, with capillary pressure determining accessibility to small pore throats and varying among gases due to differences in interfacial properties. In contrast, during the imbibition stage, the capillary pressure path differs from that of drainage, leading to hysteresis that hinders complete gas release and results in residual gas trapping at the pore scale.

Drainage corresponds to the injection phase in UHS. The entry pressure was 3.4 bar, whereas cushion gases required entry pressures of 0.8–2.4 bar, as shown in Fig. 8(a). As water saturation decreased from approximately 50%, all capillary pressure curves rapidly climbed to a plateau of 1–4 bar. Hydrogen's maximum capillary pressure reached 23.4 bar, significantly exceeding methane's peak of ~11.8 bar.

During imbibition, representing the production phase in UHS, the maximum capillary pressures decreased compared to drainage (H₂ around 14.7 bar; CH₄ around 7.9 bar), as shown in Fig. 8(b). At 50% water saturation, capillary pressures ranged from 0.5 to 2.7 bar. Subsequently, hydrogen's residual gas saturation was 5.3%, whereas methane's residual gas saturation was around 14.8%.

3.2.2. Relative permeability

Relative permeabilities can be split into the gas relative permeability and the water relative permeability, representing a reservoir's gas and water flow capacities, respectively. The gas relative permeabilities of H₂ and various cushion gases have been obtained utilizing the PNM simulations, as shown in Fig. 9(a). CO₂ has the highest flow capability, followed by N₂, CH₄, and H₂. If using the K_{rg} value of 0.5 as a standard, the water saturations for CO₂, N₂, CH₄, and H₂ are 42%, 33%, 27%, and 17%, respectively. This implies that CO₂ can spread more continuously in the pore space and thus form a more effective buffer zone, which helps suppress water-intrusion channel formation in UHS. This difference mainly arises from gas-dependent variations in flow mobility and interfacial properties under the same water saturation conditions. Furthermore, the H₂ relative permeability shows that it can only move rapidly in rocks with low water saturation, with its gas relative permeability dropping below 0.25 when water saturation is above 27.5%, which shows that considerable resistances need to be overcome at a leading edge of H₂ during an injection process.

Fig. 9(b) shows the results of the water relative permeabilities results. It appears that the differences in water relative permeability among various cushion gases are smaller than those in gas relative permeability. Water saturation for H₂ is the maximum at 75%, while it is

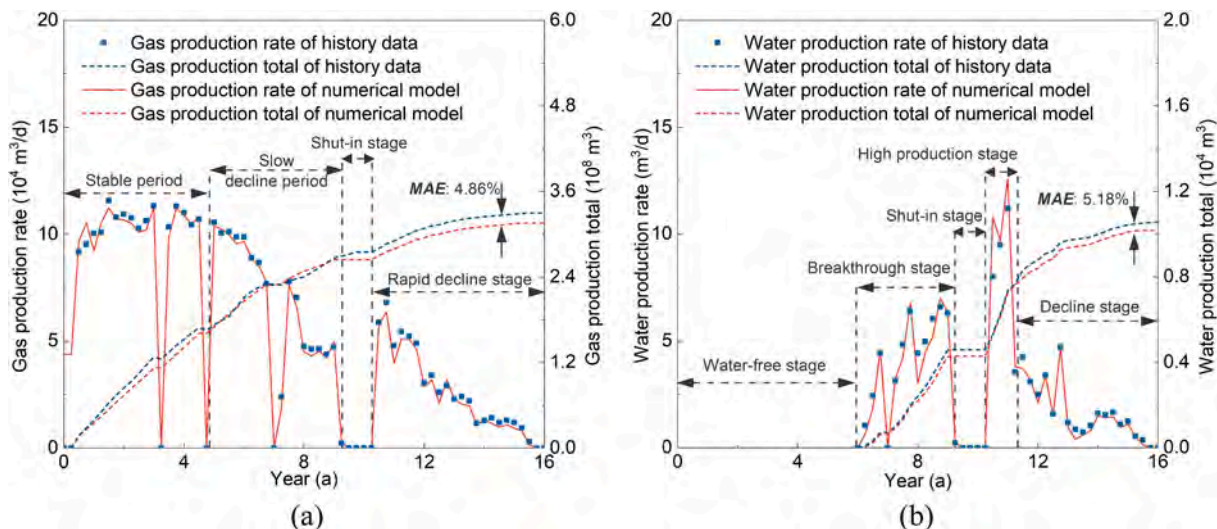


Fig. 6. Comparison of simulated and historical production data. (a) Simulated and historical gas-production curves. (b) Simulated and historical water-production curves. Where MAE is Mean Absolute Error.

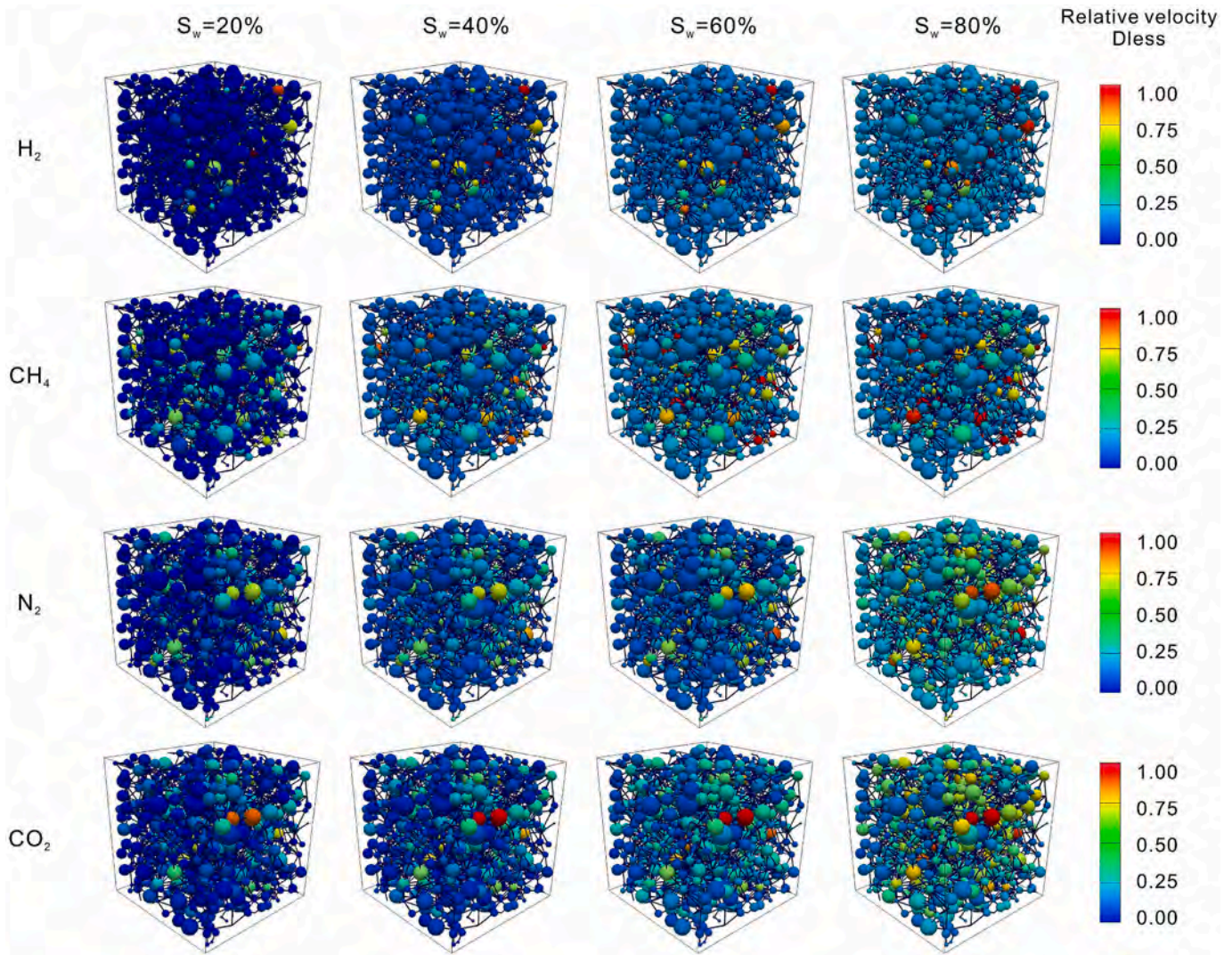


Fig. 7. The relative velocity of different gases via water saturation.

the lowest for CO₂ at 61%, with an overall difference of just 14% when using $K_{rw} = 0.5$ as a comparison value. In terms of the cycle processes of hydrogen injection and production in UHS, a slower water-phase flow speed would promote a more uniform dispersion of the injected gas across the geological layer. Consequently, CO₂ still offers advantages over other cushion gases, even though differences in water relative permeability are less apparent than those in gas relative permeability, thereby helping to reduce the velocity of the mixing zone at the CO₂–water interface. In the reservoir simulation, these pure-gas relative permeability curves were used as a basis for dynamic interpolation, and the actual relative permeability in each grid block was updated according to the local gas composition.

3.3. Hydrogen production

3.3.1. Production ability

This study used a field model of aquifer-driven gas reservoir A to test various cushion gases in cyclic UHS operations, as shown in Fig. 10. Four cycles were simulated, each with five months of injection, one month of shut-in for pressure equilibration, and six months of hydrogen production.

The production performance during the first cycle under different cushion gas scenarios is shown in Fig. 10(a) and Fig. 10(b). Hydrogen recovery in all cushion gas cases shows an initial rapid increase followed

by a decelerated growth stage, and then gradually reaches a steady state, as shown in Fig. 10(a). Among the tested cases, CO₂ achieves the highest final recovery ($\approx 79.3\%$), whereas CH₄ results in the lowest recovery ($\approx 69.4\%$). In contrast, in the case without cushion gas, the hydrogen recovery rate remains nearly unchanged due to the lack of sufficient pressure support. The water invasion volume curve also follows three distinct stages, with a major increase occurring during the final month, as shown in Fig. 10(b). Among the tested cases, CO₂ cushion gas most effectively suppresses cumulative water invasion, whereas CH₄ exhibits the weakest control.

The multi-cycle performance of hydrogen recovery and water cut is shown in Fig. 10(c) and Fig. 10(d). As shown in Fig. 10(c), the hydrogen recovery rate increases progressively with successive cycles for all cushion gas scenarios. Specifically, in the CO₂ case, the recovery rate increases from approximately 80% to about 90%, whereas the CH₄ case reaches a final recovery of around 82%. Correspondingly, the water cut decreases with cycles for all cushion gas cases, as shown in Fig. 10(d). In the CO₂ case, water cut declines from 15% to approximately 5%, showing the most effective suppression of water invasion. In contrast, in the case without cushion gas, the water cut remains consistently above 30%, as water forms high-velocity flow pathways.

The hydrogen purity under different cushion gas conditions is shown in Fig. 11(a). In the case without cushion gas, the reservoir maintains relatively high hydrogen purity throughout the production period,

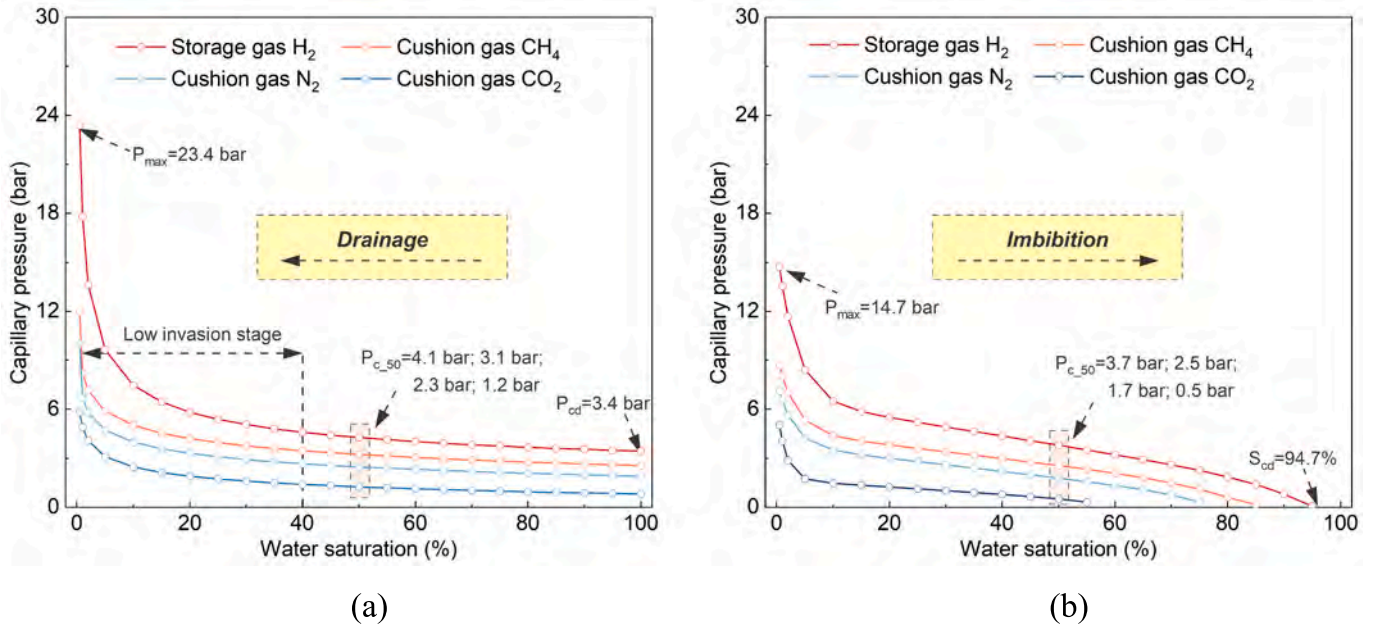


Fig. 8. Capillary pressure curves for hydrogen and cushion gases calculated using the PNM simulation method. (a) Drainage process. (b) Imbibition process.

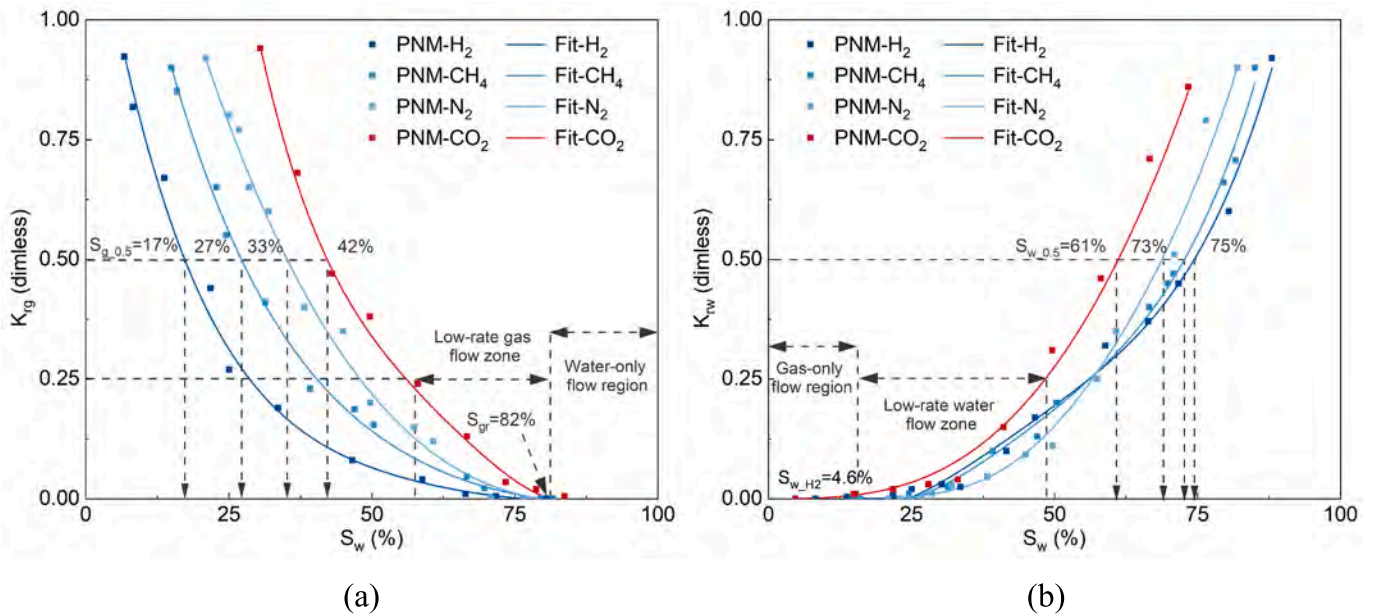


Fig. 9. Relative permeability curves of hydrogen and various cushion gases calculated with the PNM model. (a) Gas relative permeability curves. (b) Water relative permeability curves.

stabilizing at approximately 48%. In contrast, when the cushion gas is introduced, the presence of multiple gas components in the reservoir leads to dilution of hydrogen in the produced gas, resulting in a maximum reduction in hydrogen purity of up to 29.6%. The average hydrogen purity over four operational cycles is shown in Fig. 11(b). As the number of cycles increases, the average hydrogen purity exhibits a consistent upward trend across all the cases. Among the cushion gas options considered, CO₂ injection results in a relatively higher purity of approximately 66%. In contrast, in the case without cushion gas, the purity remains the highest, reaching approximately 80%.

Among the cushion gas cases, CO₂ shows the smallest decline in hydrogen purity over repeated cycles, as its higher density limits gas mixing. In contrast, the case without cushion gas yields slightly higher hydrogen purity. However, Fig. 10 shows that it significantly reduces

hydrogen recovery and increases water cut. Therefore, a cushion gas selection in UHS needs to balance hydrogen purity against overall production performance.

3.3.2. Saturation distribution

The horizontal saturation distribution of different cushion gas scenarios after four operational cycles is shown in Fig. 12. Rather than being uniformly distributed, all gases preferentially accumulate at higher elevations. For example, Fig. 12(b) highlights a red zone between Wells 3 and 4 at ≈ 26 m above the surrounding area. Saturation differences are most pronounced in lower-elevation regions, where CO₂ reaches higher saturation than other gases, especially at boundary wells (Wells 1 and 4).

Fig. 12(c) compares horizontal containment performance over four

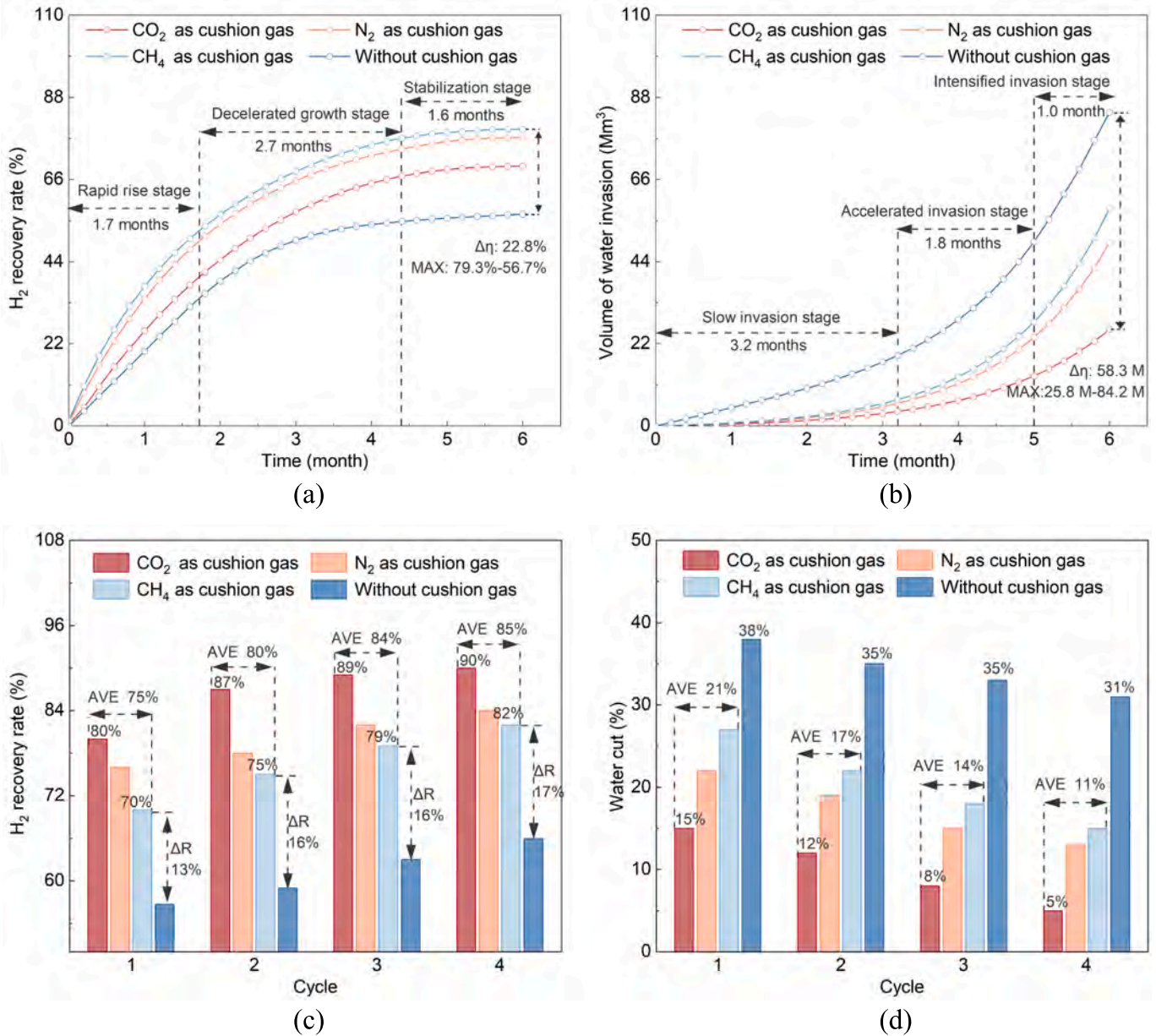


Fig. 10. Production performance of UHS under different cushion gas scenarios. (a) Hydrogen recovery rate in the first cycle; (b) Volume of water invasion in the first cycle; (c) Hydrogen recovery rate across cycles; (d) Water cut across cycles.

cycles using two metrics: (I) maximum horizontal protection distance and (II) percentage of high-saturation area. For CO₂, the maximum distance reached 614 m in cycle 1 and remained 583 m by cycle 4, while its high-saturation area decreased from 26% to 20%. In contrast, CH₄'s protection distance fell to 328 m and its high-saturation area dropped to 13% by the fourth cycle, which suggests that CO₂ maintains superior containment performance over multiple cycles, particularly in lower-elevation zones. This heterogeneity is mainly caused by the combined effects of gas-dependent density, viscosity, capillary behavior, and reservoir structure, which together control the migration and buffer-zone formation of different cushion gases. These differences indicate that a cushion gas selection should be considered together with an injection rate, well placement, and aquifer influence to improve containment performance and mitigate water intrusion.

Fig. 13 shows a vertical saturation distribution. In Fig. 13(b), after the fourth injection–production cycle, CO₂ injection forms a high-hydrogen saturation zone adjacent to the wellbore. In contrast, when CH₄, or N₂ is used as cushion gases, hydrogen flows away from the wellbore.

This pattern reflects hydrogen's high capillary pressure. When the cushion gas does not sufficiently isolate the aquifer, water invasion increases significantly due to strong capillary forces between water and hydrogen.

Fig. 13(c) quantifies vertical containment performance across cycles using two metrics: (I) the maximum vertical protection depth and (II) percentage of the high saturation zone. For CO₂, the protection depth peaked at 42 m and the high saturation volume fraction at 24% in cycle 1, then stabilized at 33 m and 20% by cycle 4. In contrast, N₂ and CH₄ showed larger declines; CH₄'s protection depth fell to 18 m and the high saturation percentage to 10.2%. These findings suggest that CO₂ provides superior aquifer isolation, particularly in the lower-elevation regions of the undulating model, where its barrier effect is most pronounced.

3.4. Economic assessment

Economic benefits are a critical factor in evaluating the feasibility of

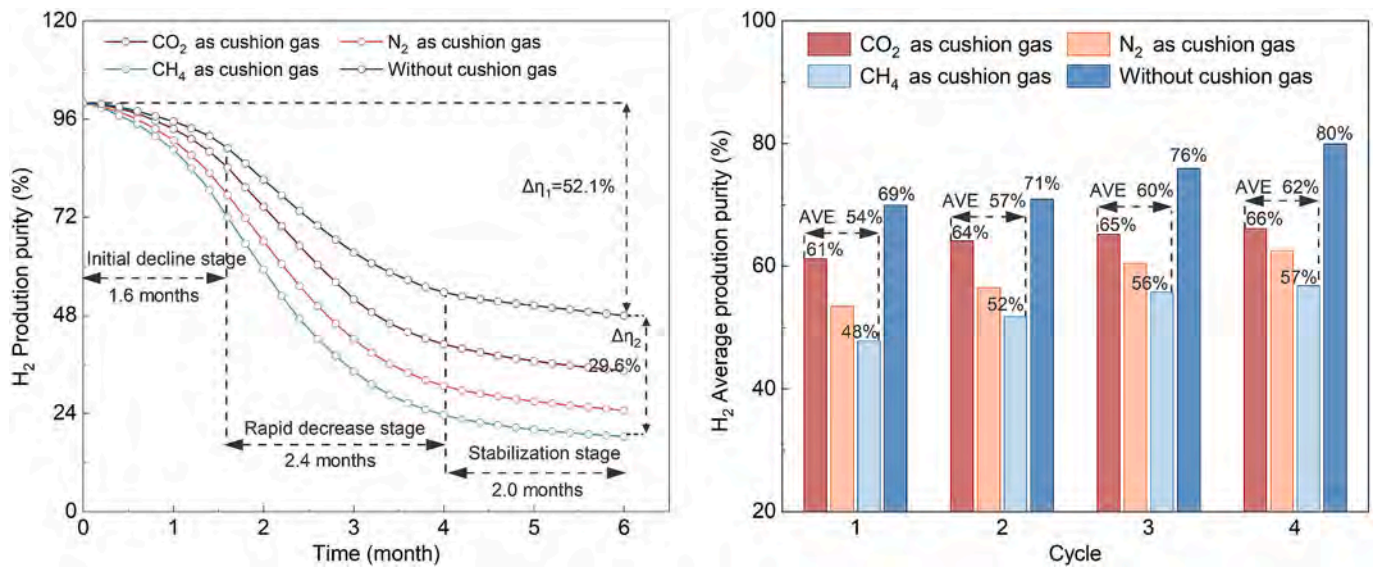


Fig. 11. Hydrogen purity in UHS under different cushion gas conditions. (a) Hydrogen purity variation during the first operation cycle; (b) Hydrogen purity variation over four operation cycles.

UHS projects. The cost parameters were based on practical data from the economic evaluation requirements for underground gas storage construction in China. Considering the high-rate injection and production demand of UHS operation, new wells were proposed, while parts of the existing surface facilities and equipment from the original gas reservoir development were reused.

3.4.1. Capital expenditure

The CAPEX considered in the economic assessment includes construction costs, cushion gas costs, existing asset value, and contingency funds, as shown in Table 2. The total CAPEX of the project ranges from USD 64.78 million to USD 82.98 million, depending on the selected cushion gas.

Regarding construction costs, including well drilling, completion, and surface engineering, these items are considered in the CAPEX estimate. These costs represent the largest expenditure category in the UHS project, amounting to USD 56.21 million and accounting for more than 77.78% of the total CAPEX.

The cushion gas cost constitutes another major component of CAPEX, with a notable variation depending on the type of cushion gas selected. This study considers the market prices of three cushion gases over the past ten years, including the maximum, average, minimum, and current prices. Among the three cushion gases, CH₄ shows the highest cost and the largest price fluctuation over the past decade. Its maximum price reaches USD 145,384.62 per MCM, while the minimum price is USD 90,574.62 per MCM, corresponding to a price variability of approximately 67.23%. Following CH₄, CO₂ exhibits a price range from USD 4.67 million to USD 11.56 million, with a price variability of about 59.60%, which is approximately 7.62% lower than that of CH₄. Similarly, the existing asset values, including factory equipment and injection equipment in the depleted gas reservoir, are estimated at USD 2.34 million and USD 1.01 million, respectively.

In addition, a contingency fund is included in the CAPEX, comprising a construction contingency fund and allowances for unforeseen operational expenditures. The construction contingency fund is estimated at USD 3.08 million and approximately 2.5% of the total CAPEX.

where \$ is USD, \$M is USD million, MCM is million cubic meters.

3.4.2. Operating expenditure

The OPEX of the UHS includes both production and indirect expenses, as summarized in Table 3. Regarding the injection fee in the

production section, equipment aging is considered. In the short term (the first eight years), the injection cost is estimated at approximately USD 3.90 million; thereafter, it is expected to increase to USD 5.70 million due to gradual equipment aging. In addition, UHS facility operation costs include hydrogen purification, labor, fuel, and other expenses, totaling USD 3.02 million. Furthermore, depreciation and amortization include equipment depreciation, amounting to USD 0.56 million. Management and financial costs are also included in the indirect expenses, amounting to USD 2.16 million and USD 1.40 million, respectively.

3.4.3. Assessment results

To assess the different profits based on market uncertainties, this study calculates the NPV using various parameters, as shown in Fig. 14. Regarding the prices of different cushion gases over the past ten years, as shown in Fig. 14(a), the different cushion gas prices have a negligible effect on the NPV profit. The best choice is CO₂, with NPV ranging from USD 51.10 million to USD 61.20 million.

In addition, the unforeseen operational expenditure is evaluated based on a proportion of CAPEX, ranging from 0% to 2.5%. As shown in Fig. 14(b), the effect of the expenditure is greater than that of cushion gas prices. In the worst-case scenario, utilizing CO₂ as the cushion gas can yield an NPV of USD 39.34 million for this project.

Given the uncertainty of the global economic market over the next 30 years, this study calculates the NPV using discount rates ranging from 8% to 14%. As shown in Fig. 14(c), the discount rate is a significant influencing factor. If the discount rate is 8%, the CO₂ and N₂ cushion gas schemes can all yield a profit of over USD 60 million. However, with an increase in the discount rate, the NPV drops sharply.

The storage price is another aspect that significantly influences the project benefits. As shown in Fig. 14(d), if the storage fees are maintained at the level of natural gas storage prices, approximately USD 115,000 per MCM, the CO₂ and N₂ schemes of UHS can achieve economic benefits exceeding USD 40 million.

For the 30-year long projects, technological challenges and improvements are also considered in this study. As shown in Fig. 14(e), when the production rate is reduced to 95% of the original level, the CO₂ and N₂ cases can yield an NPV of around USD 40 million. In contrast, if production increases to 110% of the current level, even the least favorable CH₄ cases can also achieve an NPV of approximately USD 55.32 million.

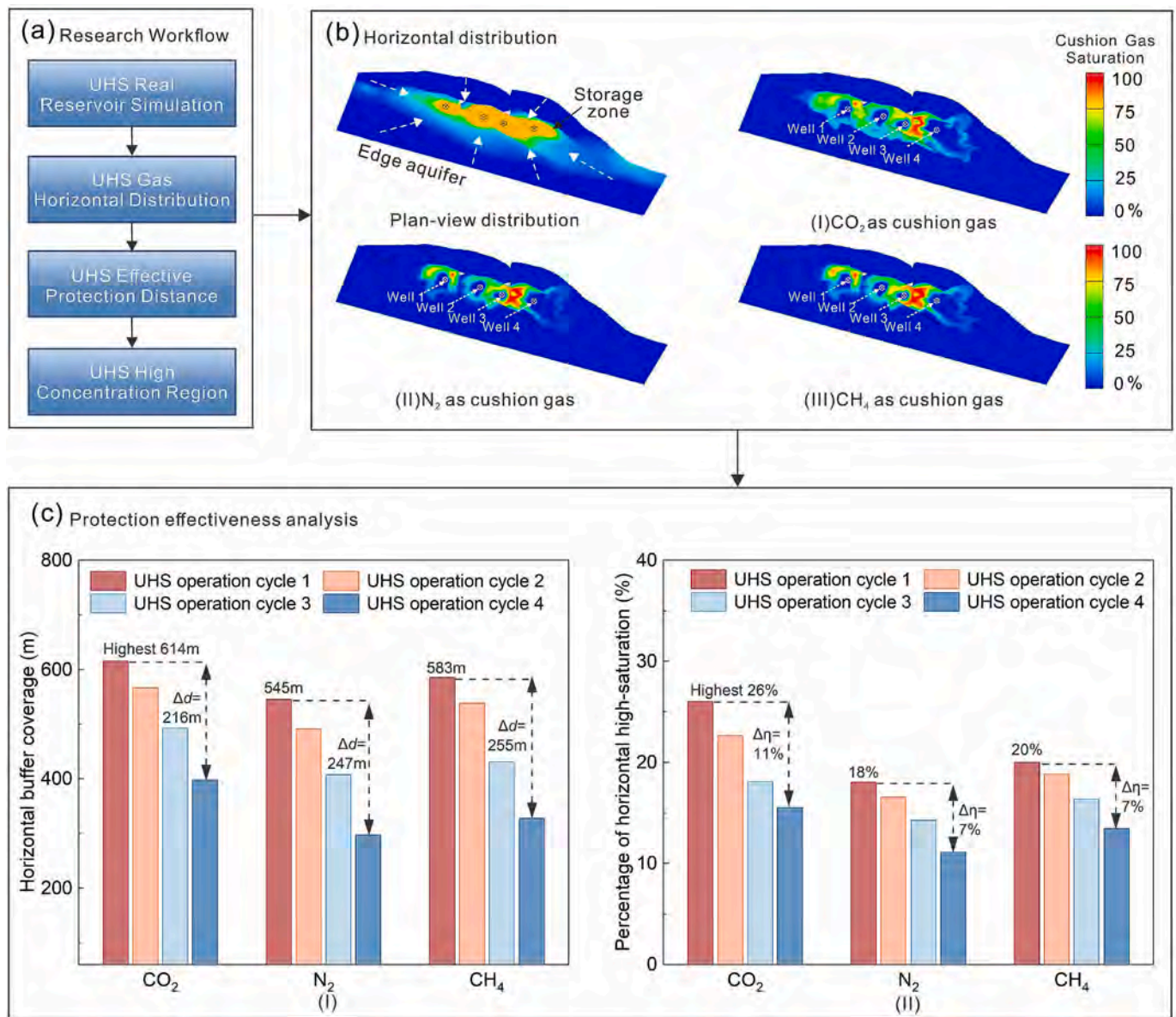


Fig. 12. Horizontal saturation distribution under different cushion gas scenarios. (a) Workflow of horizontal saturation analysis. (b) Cushion gas saturation in the fourth cycle (horizontal view). (c) Horizontal flow distribution.

The impact of gas injection fees, such as equipment aging and electricity costs, is considered in Fig. 14(f). With the injection fee ranging from USD 13,000 to 19,000 per MCM, the NPV decreases from USD 64.78 million to USD 24.07 million for the CO₂ and CH₄ cases, respectively.

To assess the performance of UHS in real-world conditions, this study analyzes the IRR for various cushion gases under three different scenarios, as shown in Table 4. The base case represents the current situation. The best case is characterized by the lowest cushion gas prices, minimal unforeseen operational expenditures, lower injection fees, and the highest level of production technology advancement. Conversely, the worst case involves the opposite conditions.

where CGP is cushion gas price, UOE is unforeseen operational expenditures, APT is advances in production technology, IF is injection fees, WT is without cushion gas, \$K is USD thousand, \$M is USD million, MCM is million cubic meters.

Except for the case without cushion gas, the UHS can remain profitable in all cushion gas cases under the current natural gas storage fee of around USD 115,000 per MCM, as shown in Fig. 15. However, there are

still differences among the various cushion gases. Due to significant fluctuations in CH₄ prices, the IRR of CH₄ exhibits a wide range from 2.8% to 14.2%, as shown in Fig. 15(b). The N₂ and CO₂ cases show good profitability, achieving IRRs of 11.63% and 13.78%, respectively, when the storage fee is set at the natural gas storage cost under the base conditions, as shown in Fig. 15(c) and Fig. 15(d).

The IRR calculations in Fig. 15 indicate the economic advantage of UHS, as the breakeven cost is USD 83,400 per MCM with the poorest performing cushion gas under the worst scenario. This study recommends using CO₂ as the cushion gas for the best financial performance and setting the hydrogen unit storage price at USD 115,000 per MCM to achieve an IRR of 13.78%. In contrast, the cost of storing H₂ in above-ground compression tanks is estimated to range from USD 1,079,137 to USD 2,068,345 per MCM. By comparison, the cost of utilizing UHS systems is approximately one-tenth of that of the above-ground storage methods. This cost difference suggests that UHS is economically competitive for large-scale hydrogen storage and may play an important role in future hydrogen energy deployment.

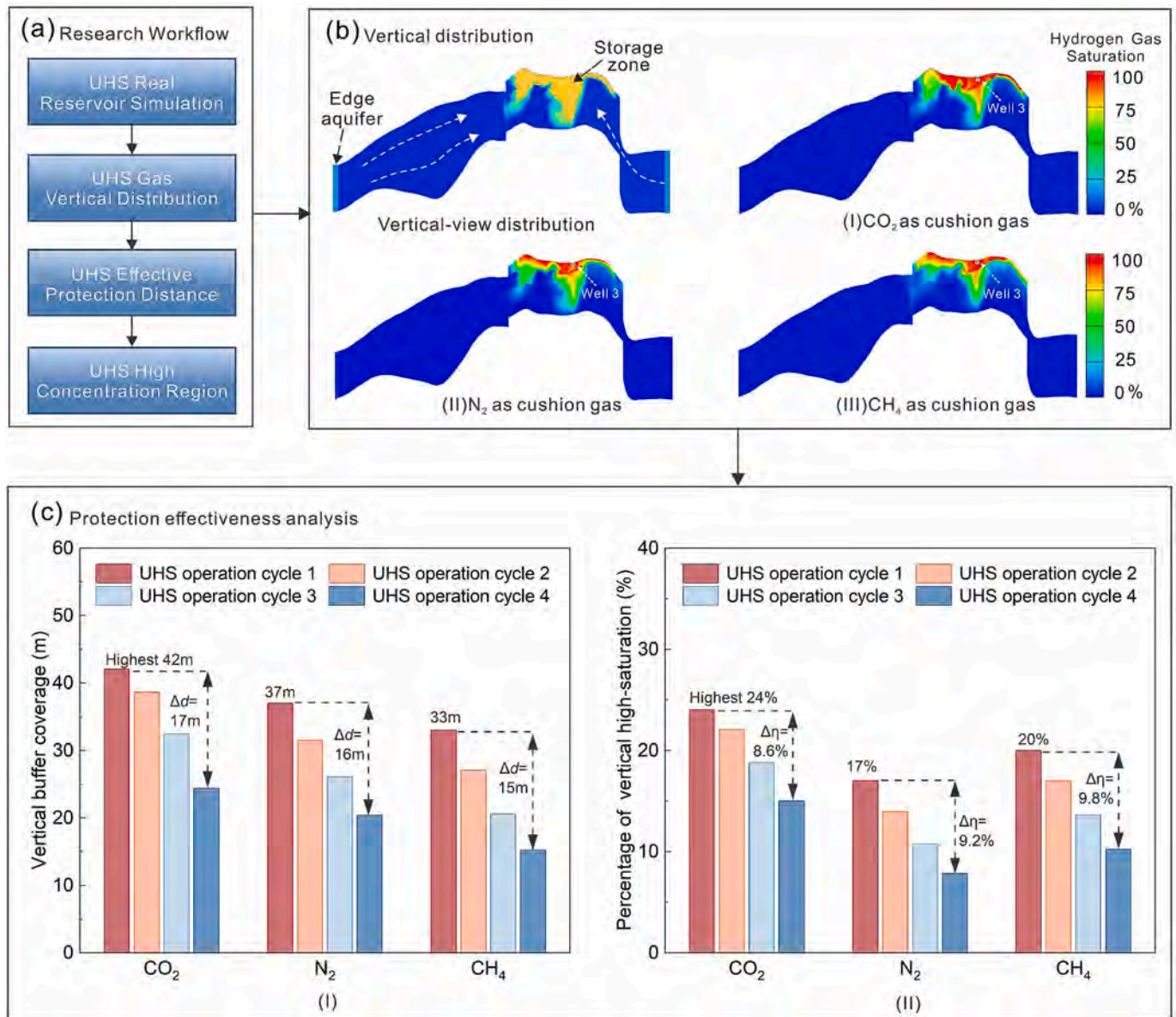


Fig. 13. Vertical view saturation distribution under different cushion gases. (a) Workflow for vertical view saturation analysis. (b) Vertical view cushion gas saturation in the fourth cycle. (c) Vertical flow distribution analysis.

4. Discussion

(1) In addition to the factors considered in this study, variations in formation water salinity may also influence the modeling results. Changes in salinity can modify gas–water interfacial tension, wettability, and gas dissolution behavior, thereby altering capillary pressure characteristics and multiphase flow processes in a reservoir.

(2) Relative permeability hysteresis may also play a role under cyclic injection–production conditions. Hysteresis can modify the distributions of residual gas and water saturations over multiple cycles, thereby influencing hydrogen recovery and water intrusion behavior. If hysteresis effects are explicitly included, the absolute recovery factors predicted by the simulations may change.

(3) The proposed PNM–numerical coupling workflow is applicable to porous-media-dominated UHS and reduces uncertainties associated with uniform relative permeability assumptions. This workflow is intended as a comparative screening tool for cushion-gas selection rather than for absolute production forecasting; since absolute predictions are reservoir-specific, the relative ranking—governed by

intrinsic gas properties—is more transferable across reservoirs.

(4) Variations in the reservoir depth influence gas flow behavior mainly through pressure- and temperature-dependent changes in gas density, viscosity, and gas–water interfacial tension, thereby affecting absolute gas mobility in porous media. With increasing the depth, these properties generally change in a similar direction for different gases and, therefore, do not alter their fundamental relative contrasts. Consequently, within the typical depth range of conventional gas reservoirs, the relative conclusions regarding water-invasion control and flow behavior among different cushion gases remain robust.

(5) Gas flow characteristics are primarily controlled by intrinsic fluid properties, including density-driven gravity segregation, viscosity-related flow resistance, and capillary behavior governed by interfacial tension. Although variations in reservoir or operating conditions may affect the magnitude of these processes, the inherent contrasts among different gases persist, resulting in stable relative rankings in flow capacity and water-isolation performance.

(6) The composition-based interpolation of relative permeability is an engineering approximation, but it follows a mature and field-

Table 2
Summary of CAPEX parameters and values.

Projects	Unit Price	Quantity	Investment
1 CONSTRUCTION			
1.1 Drilling Engineering			
· Straight wells drilling	3.46 (\$M/well)	4 Wells	\$ 13.84 M
1.2 Completion Engineering			
· Well completion	1.31 (\$M/well)	4 Wells	\$ 5.24 M
· Production equipment	0.35 (\$M/well)	4 Wells	\$ 1.40 M
1.3 Surface Engineering			
· UHS well field	4.21 (\$M/set)	1 Set	\$ 4.21 M
· 35kv power lines	0.69 (\$M/set)	1 Set	\$ 0.69 M
· Electrical substation	3.55 (\$M/unit)	1 Unit	\$ 3.55 M
· Gas gathering system	15.54 (\$M/item)	1 Item	\$ 15.54 M
· Gas transmission pipeline	8.14 (\$M/set)	1 Set	\$ 8.14 M
2 CUSHION GAS			
2.1 CH ₄			
· Current value	145,384.62 (\$/MCM)	75 MCM	\$ 10.90 M
· Maximum value	276,376.15 (\$/MCM)	75 MCM	\$ 20.73 M
· Average value	183,475.38 (\$/MCM)	75 MCM	\$ 13.76 M
· Minimum value	90,574.62 (\$/MCM)	75 MCM	\$ 6.79 M
2.2 N ₂			
· Current value	47,692.31 (\$/MCM)	75 MCM	\$ 3.58 M
· Maximum value	89,136.92 (\$/MCM)	75 MCM	\$ 6.68 M
· Average value	62,667.69 (\$/MCM)	75 MCM	\$ 4.70 M
· Minimum value	36,198.46 (\$/MCM)	75 MCM	\$ 2.71 M
2.3 CO ₂			
· Current value	86,153.85 (\$/MCM)	75 MCM	\$ 6.46 M
· Maximum value	154,160.01 (\$/MCM)	75 MCM	\$ 11.56 M
· Average value	118,181.54 (\$/MCM)	75 MCM	\$ 8.86 M
· Minimum value	62,203.08 (\$/MCM)	75 MCM	\$ 4.67 M
2.4 Injection	23,815.38 (\$/MCM)	75 MCM	\$ 1.78 M
3 EXISTING ASSET VALUE			
· Factory equipment	2.34 (\$M/item)	1 Item	\$ 2.34 M
· Injection equipment	1.01 (\$M/item)	1 Item	\$ 1.01 M
4 CONTINGENCY FUND			
4.1 Construction			
· Construction contingency fund	3.08 (\$M)	/	\$ 3.08 M
4.2 Unforeseen Operational Expenditures			
· CH ₄ as cushion gas	1.79 (\$M)	/	\$ 1.79 M
· N ₂ as cushion gas	1.61 (\$M)	/	\$ 1.61 M
· CO ₂ as cushion gas	1.68 (\$M)	/	\$ 1.68 M
TOTAL INVESTMENT			\$ 64.78–82.98 M

validated simulation practice. Although experimental relative permeability data for gas mixtures are still lacking, this treatment reflects the actual flow behavior more realistically than the conventional approach that directly adopts air–water relative permeability curves.

5. Conclusions

This study utilized PNM and numerical simulations to screen and rank various cushion gases by their relative performance in aquifer-driven UHS systems. In addition, it conducted a comprehensive economic assessment to determine the costs of the underground storage method. The main conclusions reached are as follows:

(1) Hydrogen's elevated interfacial tension makes UHS facilities susceptible to water intrusion, potentially compromising storage

Table 3
Summary of OPEX parameters and values.

Projects	Unit Price	Quantity	Investment
1 PRODUCTION			
1.1 Hydrogen Injection			
· Short term expenses	13,000.00 (\$/MCM)	300 MCM	\$ 3.90 M
· Long term expenses	19,000.00 (\$/MCM)	300 MCM	\$ 5.70 M
1.2 UHS Facility Operation			
· Hydrogen purification	3500.00 (\$/MCM)	300 MCM	\$ 1.05 M
· Fuel	1218.46 (\$/MCM)	300 MCM	\$ 0.37 M
· Labor	38,000.00 (\$/Person)	20 Person	\$ 0.76 M
· Electric	10,400.00 (\$/Well)	4 Wells	\$ 0.04 M
· Material	12,523.08 (\$/Well)	4 Wells	\$ 0.05 M
· Monitoring	40,646.15 (\$/Well)	4 Wells	\$ 0.16 M
· Wells operations	68,769.23 (\$/Well)	4 Wells	\$ 0.28 M
· Wells management	27,030.76 (\$/Well)	4 Wells	\$ 0.11 M
· Maintenance & repair	13,076.92 (\$/Well)	4 Wells	\$ 0.05 M
· Others	36,600.11 (\$/Well)	4 Wells	\$ 0.15 M
1.3 Depreciation and Amortization			
· Equipment depreciation	1875.07 (\$/MCM)	300 MCM	\$ 0.56 M
2 INDIRECT EXPENSES			
2.1 Management			
· Safety oroduction	5384.61 (\$/MCM)	300 MCM	\$ 1.62 M
· Other management	1815.38 (\$/MCM)	300 MCM	\$ 0.54 M
2.2 Financial and Sales			
· Sales	2400.00 (\$/MCM)	300 MCM	\$ 0.72 M
· Financial	2276.92 (\$/MCM)	300 MCM	\$ 0.68 M
TOTAL INVESTMENT			\$ 11.04–12.84 M

integrity. Therefore, the injection of a cushion gas is necessary to mitigate this risk.

(2) Differences in relative permeability and capillary pressure among gases strongly influence flow behavior in UHS. Accurate UHS numerical simulation requires distinguishing these differences to avoid notable errors in selecting cushion gases.

(3) CO₂ demonstrated better mobility and lower capillary pressure compared to N₂ and CH₄. Utilizing CO₂ as the cushion gas in UHS leads to a higher hydrogen production rate and a reduced water cut. The maximum daily hydrogen production capacity can be increased by 35.6% compared to CH₄.

(4) UHS can achieve an IRR of 13.78% with a storage price of USD 115,000 per MCM, compared to USD 1,079,137 per MCM for above-ground compression tanks. The cost of UHS is merely a tenth of that by the traditional method, making it a key factor in lowering future hydrogen storage costs and supporting the transition to a hydrogen energy society.

CRediT authorship contribution statement

Peng Deng: Writing – original draft, Methodology. **Zhangxin Chen:** Supervision. **Xiaolong Peng:** Supervision. **Chaojie Di:** Resources, Formal analysis. **Haoming Ma:** Validation, Data curation. **Chaowen Wang:** Software, Resources. **Zhenqian Xue:** Validation, Data curation.

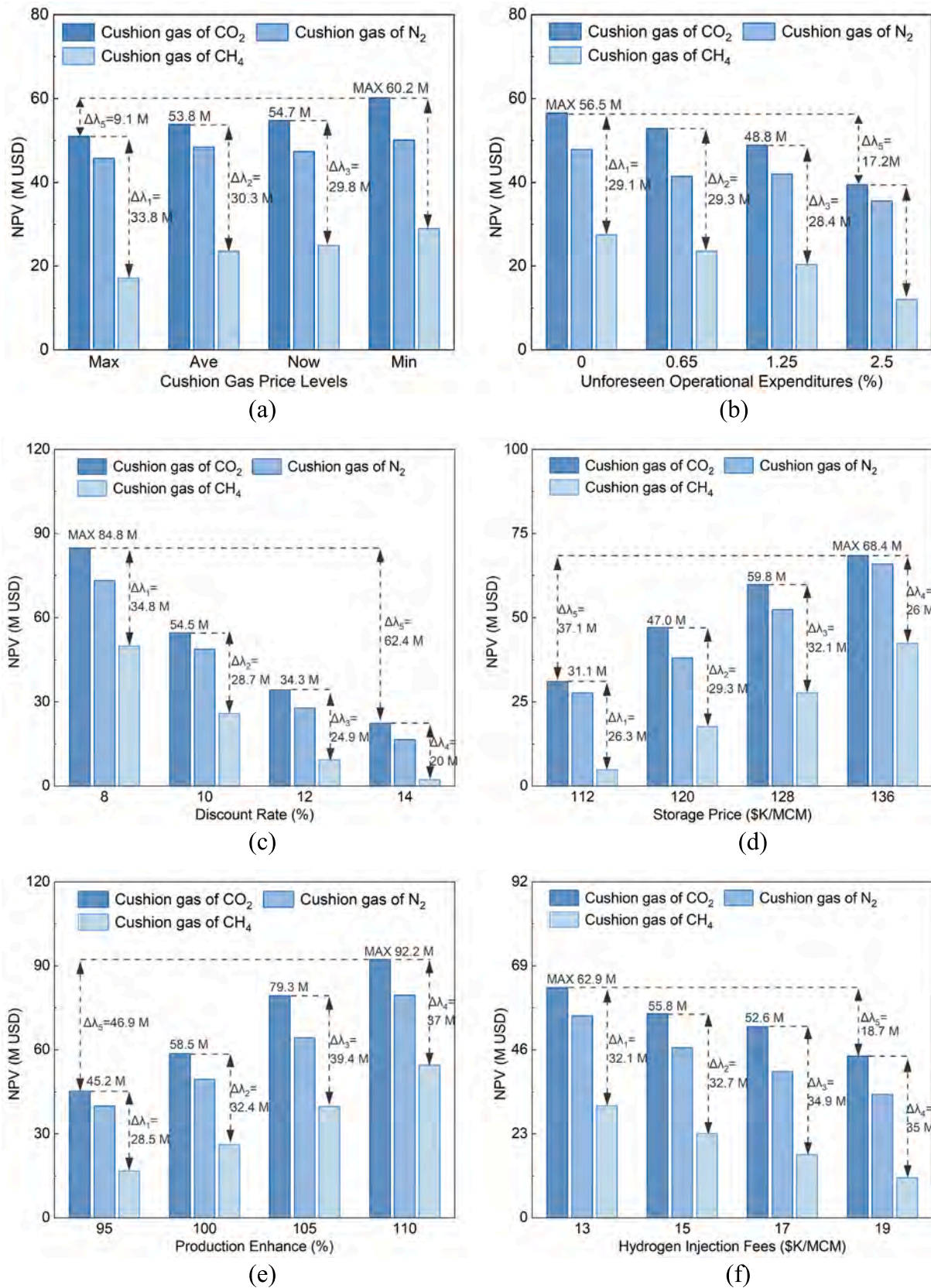


Fig. 14. NPV calculation results of the UHS with different cushion gas cases. (a) Cushion gas price level. (b) Unforeseen expenditures. (c) Discount rate. (d) Storage price. (e) Production enhance. (f) Gas injection fees.

Table 4

Three different economic evaluation models.

	CGP, \$K/MCM				UOE, \$M/a				APT %	IF \$/MCM
	WT	CH ₄	N ₂	CO ₂	WT	CH ₄	N ₂	CO ₂		
Best Case	0	94.62	37.82	64.98	0.28	0.36	0.32	0.34	110	13
Base Case	0	15.19	49.82	90.01	0.59	0.72	0.64	0.67	100	15
Worst Case	0	28.87	93.12	18.19	1.12	1.79	1.61	1.68	90	17

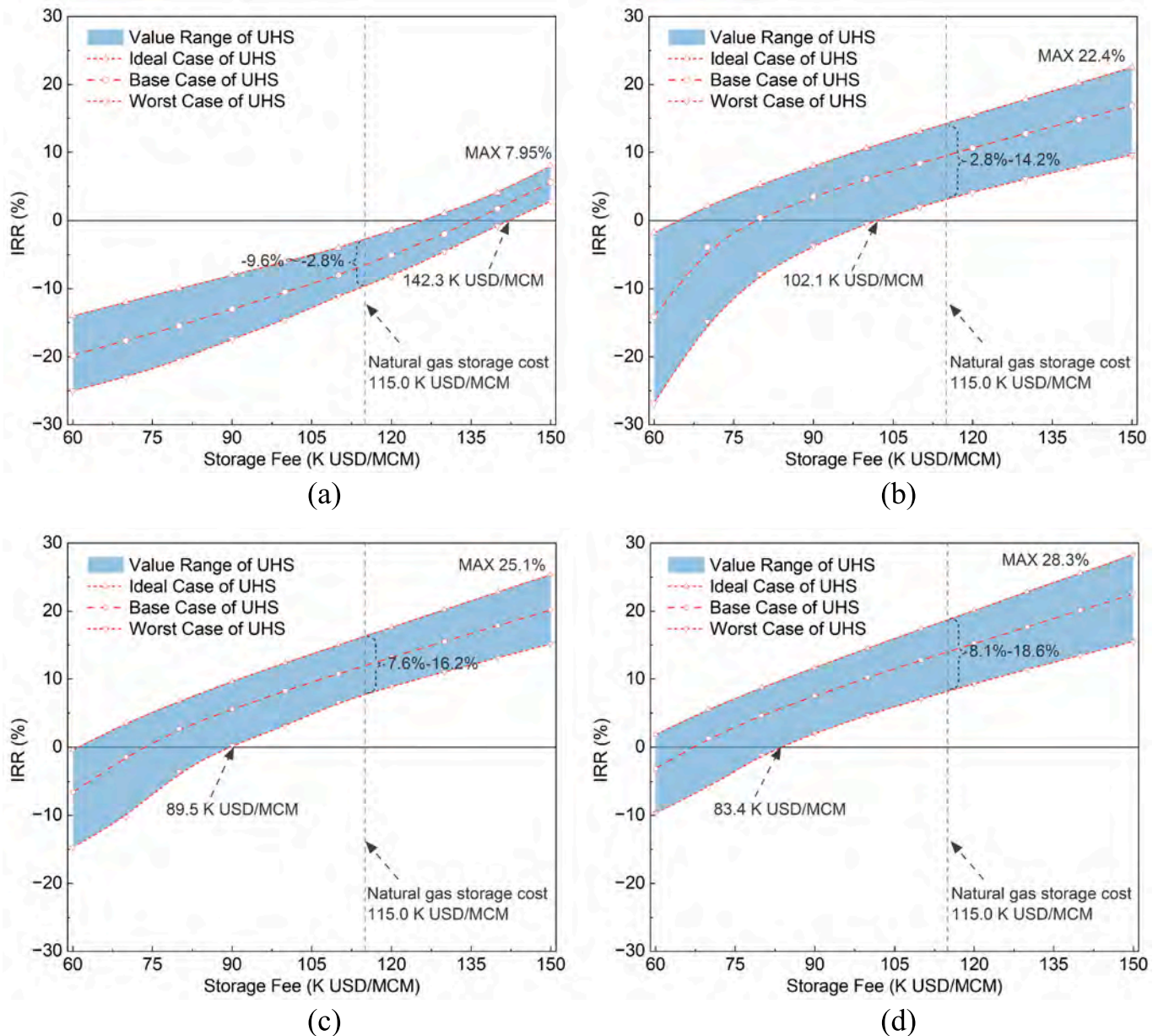


Fig. 15. IRR calculation results for the UHS under different scenarios and ranges of variation. The dashed line represents the current price of underground natural gas storage. (a) Without cushion gas. (b) CH₄ as cushion gas. (c) N₂ as cushion gas. (d) CO₂ as cushion gas.

Long Peng: Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of competing interest

interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare that they have no known competing financial

Appendix A. Appendix

List of Symbols.	
a	EOS attraction parameter [$\text{Pa}\cdot\text{m}^6\cdot\text{mol}^{-2}$]
b	EOS covolume parameter [$\text{Pa}\cdot\text{m}^6\cdot\text{mol}^{-2}$]
$(aa)_m$	Mixture attraction parameter [$\text{Pa}\cdot\text{m}^6\cdot\text{mol}^{-2}$]
C	Cost [USD]
F	Geometrical size factor [m^4]
g	Gravitational acceleration [$9.8\text{ m}\cdot\text{s}^{-2}$]
k	Permeability [m^2]
kr	Relative permeability [dimensionless]
L	Length [m]
m	Number of components [dimensionless]
MW	Molecular weight [$\text{kg}\cdot\text{mol}^{-1}$]
N	Evaluation period [a]
μ	Dynamic viscosity [$\text{Pa}\cdot\text{s}$]
μ^*	Reference liquid viscosity [$\text{Pa}\cdot\text{s}$]
n	Number of components [dimensionless]
NPV	Net present value [USD]
p	Pressure [Pa]
Q	Flow rate [$\text{m}^3\cdot\text{s}^{-1}$]
r	Radius [m]
R	Effective radius [m]
r_d	Discount rate [%]
s	Saturation [dimensionless]
T	Temperature [K]
t	Time (year index) [a]
u	Darcy velocity [$\text{m}\cdot\text{s}^{-1}$]
V	Molar volume [$\text{m}^3\cdot\text{mol}^{-1}$]
x	Mole fraction in liquid phase [dimensionless]
y	Mole fraction in gas phase [dimensionless]
Z	Vertical depth (positive upward) [m]
β	Vapor-phase fraction [dimensionless]
γ_g	Hydraulic conductance [$\text{m}^3\cdot\text{Pa}^{-1}\cdot\text{s}^{-1}$]
δ	Binary interaction coefficient [dimensionless]
λ	Empirical viscosity correction parameter [dimensionless]
ξ	Molar density [$\text{mol}\cdot\text{m}^{-3}$]
ρ	Mass density [$\text{kg}\cdot\text{m}^{-3}$]
ρ_r	Reduced liquid density [dimensionless]
φ	Porosity [dimensionless]
Subscripts.	
abs	Absolute (single-phase condition)
c	Capillary
e	Effective (multiphase condition)
g	Gas phase
h	Hydraulic (conductance)
i	Component index
I, J	Pore index
$I-J$	Connection between pores I and J
l	Liquid phase
m	Mixture
nw	Non-wetting phase
r	Relative
tax	Tax
w	Water phase

Data availability

Data will be made available on request.

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